

## A Heuristic Rule Inventories

Tables 1 and 2 list the heuristic rules used to prioritize clusters for manual review as described in Section ?? . These rules are not deployed as a standalone classifier. Instead, they serve as an empirical prioritization aid that ranks clusters by suspiciousness so that analysts can focus their effort on the most promising candidates first. Many rules target indicators that are well established in the malicious JavaScript detection literature, including AST-level obfuscation signatures and suspicious API calls identified by Zozzle [?] and JaSt [?], page-level entropy and packing anomalies used by Prophiler [?] and Cujo [?], and runtime API-call patterns studied by WeBehavior [?]. Unlike those systems, which train classifiers to label individual scripts, our rules rank clusters for manual review. Beyond these broadly shared indicators, 15 rules (12 static and 3 behavioral, marked with  $\diamond$  in Tables 1 and 2) draw specifically on the obfuscation taxonomy of Zhou et al. [?]. Domain-specific rules were seeded from public threat intelligence reports on WordPress injection campaigns [???], and the remaining rules emerged from manual source-code analysis over the 24 crawl windows.

**Table 1: Static JavaScript heuristic rules (37 rules). Rules operate on raw source code and are organized by category. Wt.: rule weight. Campaign coverage notes indicate how many of the 20 confirmed campaigns trigger each rule.  $\diamond$  Zhou et al. [?],  $^Z$  Zozzle [?],  $^J$  JaSt [?],  $^P$  Prophiler [?],  $^C$  Cujo [?]. A superscript marks prior work that uses the same JavaScript construct as a classification feature, and it does not mean that the cited work defines our rule. Zozzle and JaSt, for instance, are syntactic classifiers that specify no deobfuscation rules, yet both use constructs such as `eval`, `String.fromCharCode`, and hexadecimal variable names as features.**

| Category               | Rule ID                              | Wt. | Description                                                                       |
|------------------------|--------------------------------------|-----|-----------------------------------------------------------------------------------|
| Obfuscation Signatures | obf_0x_naming $\diamond^Z$           | 3.0 | <code>_0x[a-f0-9]+</code> variable names ( $\geq 5$ occurrences; 14/20 campaigns) |
|                        | obf_string_rotation $\diamond^J$     | 4.0 | String array push/shift in <code>while(![])</code> loop                           |
|                        | obf_JSFiretruck $\diamond$           | 5.0 | JSFiretruck encoding ( $>90\%$ of chars are <code>[]()</code> !+; JSFiretruck)    |
|                        | obf_hex_escapes $\diamond^Z$         | 3.0 | Bulk <code>\x..</code> sequences ( $\geq 20$ ; CookieLoader, OverlayInject)       |
|                        | obf_xor_decrypt $\diamond^Z$         | 4.5 | <code>charCodeAt()</code> with <code>eval/new Function</code> (PolyClickFix)      |
|                        | obf_custom_b64_alphabet $\diamond$   | 2.5 | Full base64 alphabet as string literal (HexArray, GDPRInject)                     |
|                        | obf_large_b64_blob                   | 3.0 | Base64 string $\geq 500$ chars                                                    |
|                        | obf_bracket_prop_access $\diamond^J$ | 2.0 | <code>[_0x... (0x...)]</code> property access ( $\geq 5$ calls)                   |
|                        | obf_parseint_checksum $\diamond$     | 2.0 | <code>parseInt(_0x...)</code> checksum loop ( $\geq 4$ calls)                     |
|                        | obf_string_reverse $\diamond^J$      | 2.5 | <code>split("").reverse().join("")</code> for API-name hiding                     |
|                        | obf_fromcharcode_loop $\diamond^Z$   | 2.5 | <code>String.fromCharCode</code> ( $\geq 3$ occurrences)                          |
| Suspicious API Calls   | api_eval_function $\diamond^ZC$      | 4.0 | <code>eval()</code> or <code>new Function()</code>                                |
|                        | api_create_script $^ZC$              | 3.0 | <code>createElement('script')</code> (15/20 campaigns)                            |
|                        | api_document_write $^ZC$             | 3.5 | <code>document.write</code> with script/unescape content                          |
|                        | api_atob $\diamond$                  | 2.5 | <code>atob()</code> with inline base64 (5/20 campaigns)                           |
|                        | api_cookie_manipulation              | 2.0 | Cookie setting with expiry calculation                                            |
|                        | api_sendbeacon                       | 3.0 | <code>navigator.sendBeacon()</code> (TagInject)                                   |
|                        | api_insertbefore                     | 1.5 | <code>insertBefore</code> with script creation                                    |
|                        | api_innerhtml_write $^C$             | 2.0 | <code>innerHTML +=</code> with HTML/script tags                                   |
| Evasion Patterns       | evasion_wp_cookie_check              | 5.0 | WordPress admin cookie detection (MultiC2)                                        |
|                        | evasion_bot_ua_check                 | 4.0 | Bot/crawler user-agent detection                                                  |
|                        | evasion_wp_path_filter               | 3.0 | <code>wp-admin/wp-login</code> path exclusions ( $\geq 3$ )                       |
|                        | evasion_session_gate                 | 3.5 | sessionStorage single-execution gate                                              |
|                        | evasion_anti_debug                   | 4.0 | ReDoS or constructor-based anti-debugging                                         |
|                        | evasion_timing_check                 | 2.0 | <code>performance.now()</code> timing checks                                      |
| DOM                    | dom_css_fade                         | 3.5 | <code>@keyframes fadeIn</code> CSS (SessHijack, FadeRedirect)                     |
|                        | dom_hidden_iframe $^PC$              | 3.0 | Hidden iframe ( <code>display:none</code> , <code>visibility:hidden</code> )      |
|                        | dom_overlay_inject $^C$              | 2.5 | Overlay/layer HTML injection                                                      |
| Network & C2           | net_multi_c2                         | 5.0 | $\geq 3$ base64-encoded URLs (MultiC2)                                            |
|                        | net_exfil_fingerprint                | 3.0 | Fingerprint fields + network exfiltration                                         |
|                        | net_data_param_exfil                 | 3.0 | <code>JSON.stringify + encodeURIComponent + ?data=</code>                         |
|                        | net_suspicious_domain                | 4.0 | Known malicious domains (13 IOC domains)                                          |
| WP                     | wp_rogue_user                        | 5.0 | Rogue user creation (WooBackdoor)                                                 |
|                        | wp_rest_api_abuse                    | 3.0 | <code>/wp-json/</code> with credential keywords                                   |
| Size/Entropy           | anomaly_single_line $^P$             | 2.0 | $\geq 2,000$ bytes on $\leq 3$ lines                                              |
|                        | anomaly_high_entropy $^JP$           | 2.0 | Shannon entropy $>5.5$ bits/char                                                  |
|                        | anomaly_low_whitespace $^J$          | 1.5 | Whitespace ratio $<2\%$ (packed code)                                             |

## B CDP Instrumentation Hooks

Table 3 enumerates all 76 browser APIs instrumented by the behavioral monitoring script injected via `Page.addScriptToEvaluateOnNewDocument` (Section ??).

We compiled this list from three sources. First, we adopted the instrumentation surfaces of prior in-browser monitoring and drive-by detection systems, namely VisibleV8 [?], WeBehavior [?], and Cujo [?]. Second, we added the APIs that public WordPress threat intelligence

**Table 2: Behavioral trace heuristic rules (33 rules). Rules operate on CDP execution traces and incorporate baseline subtraction to discount monitoring-harness overhead (Section ??). <sup>◦</sup> Zhou et al. [? ], <sup>Z</sup> Zozzle [? ], <sup>P</sup> Prophiler [? ], <sup>C</sup> Cujo [? ], <sup>W</sup> WEBBEHAVIOR [? ]. Superscripts carry the same meaning as in Table 1 and mark prior work that treats the same construct or API call as a classification feature.**

| Category                    | Rule ID                                    | Wt. | Description                                                 |
|-----------------------------|--------------------------------------------|-----|-------------------------------------------------------------|
| Network Communication       | net_xhr_external <sup>CW</sup>             | 3.0 | XHR requests to external domains                            |
|                             | net_xhr_multi_domain                       | 2.0 | XHR to $\geq 3$ distinct domains (C2 fallback pattern)      |
|                             | net_fetch_external <sup>CW</sup>           | 3.0 | Fetch API requests to external domains                      |
|                             | net_beacon <sup>W</sup>                    | 3.0 | sendBeacon API call for exfiltration                        |
|                             | net_script_src_external <sup>C</sup>       | 2.0 | External script injection via .src property                 |
|                             | net_script_src_php                         | 2.0 | Script src to .php endpoint with data parameter             |
|                             | net_suspicious_tld                         | 1.0 | Communication with suspicious TLDs (.cf, .shop, .top, .biz) |
|                             | net_tds_params                             | 2.0 | TDS-style tracking parameters (se_referrer, landing_url)    |
|                             | net_blockchain_rpc                         | 3.0 | Blockchain RPC endpoint (Polygon/Ethereum; PolyClickFix)    |
|                             | net_known_malicious                        | 4.0 | Known IOC domains (17 hardcoded campaign C2s)               |
| Code Injection              | inject_docwrite_iframe <sup>PC</sup>       | 4.0 | document.write with iframe (JSFiretruck)                    |
|                             | inject_docwrite_script <sup>C</sup>        | 3.0 | document.write with external script (CookieLoader)          |
|                             | inject_docwrite_css_hide                   | 3.0 | document.write with CSS opacity/z-index hiding              |
|                             | inject_docwrite_multiple                   | 2.0 | $\geq 3$ document.write calls                               |
|                             | inject_eval <sup>◦ZCW</sup>                | 2.0 | eval() calls at runtime                                     |
|                             | inject_function_constructor <sup>◦CW</sup> | 2.0 | new Function() constructor invocation                       |
| Finger-printing             | inject_atob_deobfuscation <sup>◦</sup>     | 1.5 | atob() base64 decoding at runtime                           |
|                             | fp_heavy_prop_read <sup>W</sup>            | 2.0 | DOM property reads $>30$ above baseline                     |
|                             | fp_hardware_fingerprint                    | 2.0 | hardwareConcurrency, deviceMemory enumeration               |
|                             | fp_webgl                                   | 2.0 | WebGL context creation                                      |
|                             | fp_worker_blob                             | 1.5 | Web Worker + Blob URL combination                           |
|                             | fp_useragent_repeated                      | 1.0 | navigator.userAgent read $\geq 5$ times                     |
| Event Hooking & Persistence | hook_excessive_listeners                   | 2.0 | Event listeners $>10$ above baseline                        |
|                             | hook_interaction_hijack                    | 2.0 | Click/mouse/keyboard hooks on document                      |
|                             | hook_ad_network_events                     | 2.0 | Ad-network keywords (impression, missclick, adcash)         |
|                             | hook_excessive_defineprop                  | 1.0 | defineProperty $>15$ above baseline                         |
|                             | hook_nonstandard_defineprop                | 1.5 | defineProperty on suspicious targets                        |
|                             | hook_timeout_cascade                       | 1.0 | setTimeout $>10$ above baseline                             |
|                             | hook_interval_persistent                   | 1.0 | setInterval usage                                           |
| Exfil.                      | hook_visibility_persistence                | 1.0 | visibilitychange/pageshow listeners ( $\geq 2$ types)       |
|                             | exfil_cookie_read <sup>W</sup>             | 1.5 | Cookie read combined with network activity                  |
|                             | exfil_cookie_gate                          | 1.5 | Cookie write combined with script injection                 |
|                             | exfil_postmessage                          | 1.5 | postMessage cross-origin communication                      |

reporting names in live injection campaigns, drawing on Sucuri [? ], Rapid7 [? ], and Palo Alto Networks Unit 42 [? ? ]. Third, we manually analyzed the payloads captured during the first three crawl windows and added the APIs those samples exercised that neither of the first two sources covered. The first two sources give coverage of behaviors already known to the literature and to industry, and the third closes the gap toward the payloads that our own measurement observed in the wild.

## C Campaign Catalog

Table 4 provides the full catalog of the 20 canonical campaigns introduced in Section ??, listing the fingerprintable indicators extracted by the pipeline and the behavior each campaign produces on victims and site operators. Campaign variants whose injection, C2 infrastructure, and behavior are otherwise identical are merged into a single row (for example, HexArray-A/B/C and GDPRInject).

## D Heuristic Rule Evaluation

The heuristic rules serve as a cluster prioritization aid rather than a standalone classifier (Section ??). To characterize their individual specificity, we evaluated all 70 rules against a human-verified ground truth corpus of 5,000 unique benign JavaScript files constructed during the study. Each file was drawn from WordPress sites whose crawl snapshots contained no injected deltas across consecutive windows and whose page content was manually confirmed as benign, then deduplicated by SHA-256 so that widely deployed libraries appear only once. We evaluated both the static source code and the corresponding execution trace of each file, so the static and behavioral columns in Table 5 reflect the same population of scripts observed under both analysis modalities. Table 5 reports the results against this corpus and the 61 confirmed malicious campaign payloads.

Static rules exhibit a higher false positive rate (22.7%) because legitimate WordPress JavaScript routinely triggers low-weight indicators such as minified single-line code (48% of benign files), insertBefore calls (26%), and createElement('script') (20%). Because these patterns appear in nearly half of all benign files, they receive low individual weights (1.0 to 2.0) so that no single pattern alone can accumulate sufficient score to cross the detection threshold. Behavioral rules achieve a false positive rate of 0.06% (3 out of 5,000) because benign scripts

**Table 3: All 76 browser APIs instrumented by the behavioral monitoring script, grouped into 14 categories. Each interposed API emits a structured JSON event containing the event type and a type-specific subset of parameters.**

| Category            | Hooked APIs                                                                                                                                                                                                                            | #  | Notes                                                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------------------------------------------------------------------------------------------------------------------------------|
| Network I/O         | XMLHttpRequest (open, setRequestHeader, send, response), fetch, WebSocket (constructor, send, onmessage), navigator.sendBeacon                                                                                                         | 7  | XHR lifecycle, fetch request+response, Web-Socket messaging, beacon exfiltration                                             |
| DOM mutation        | appendChild, removeChild, replaceChild, insertBefore, removeAttribute, toggleAttribute, setAttribute, DOMTokenList.add/remove/toggle, CSSStyleDeclaration.setProperty, innerHTML, outerHTML (setter), document.write, document.writeln | 15 | Child-node, attribute, class-list, and style mutation; HTML property writes with script detection; document stream injection |
| Script injection    | document.createElement, eval(), Function() constructor                                                                                                                                                                                 | 3  | Hooks script.src, script.textContent, iframe creation; dynamic code evaluation                                               |
| Navigation          | location.href (setter), location.replace, location.assign, history.pushState, history.replaceState, window.open, HTMLFormElement.submit                                                                                                | 7  | Location redirects, History API, popup/tab navigation, programmatic form submission                                          |
| Storage             | document.cookie (get+set), storage event, localStorage, sessionStorage, indexedDB, caches                                                                                                                                              | 5  | Cookie read/write, cross-tab storage events, property interceptors                                                           |
| Timers              | setTimeout, setInterval                                                                                                                                                                                                                | 2  | Deferred and periodic execution                                                                                              |
| De-obfuscation      | atob(), String.fromCharCode(), JSON.parse                                                                                                                                                                                              | 3  | Base64 decoding, char-code assembly, suspicious payload parsing                                                              |
| Sensitive data      | HTMLInputElement.value, HTMLTextAreaElement.value, HTMLSelectElement.value (get+set), clipboard.writeText, clipboard.readText, geolocation.getCurrentPosition, geolocation.watchPosition, querySelector, getElementById                | 9  | Form fields filtered for password/payment; clipboard and geolocation access; credential-selector queries                     |
| Workers / messaging | Worker (constructor, postMessage, onmessage), serviceWorker.register, BroadcastChannel, window.postMessage, message event                                                                                                              | 6  | Web Worker lifecycle, service worker registration, cross-tab and cross-origin messaging                                      |
| Observers           | IntersectionObserver, ResizeObserver, MutationObserver                                                                                                                                                                                 | 3  | Viewport visibility, element resize, native backup for attribute/child-list changes                                          |
| Evasion             | Object.defineProperty, Function.prototype.toString, IFrame creation, Element.attachShadow                                                                                                                                              | 4  | Prototype pollution detection, fake [native code] return, iframe alert when > 3, shadow DOM                                  |
| UI manipulation     | window.open, Element.requestFullscreen, alert/confirm/prompt, URL.createObjectURL, HTMLAnchorElement.click                                                                                                                             | 5  | Popups, fullscreen overlay, dialog interception, blob URL creation, triggered downloads                                      |
| Canvas / push       | HTMLCanvasElement.getContext, HTMLCanvasElement.toDataURL, PushManager.subscribe                                                                                                                                                       | 3  | Canvas fingerprinting, data export, push notification subscription                                                           |
| Error / lifecycle   | window.onerror, window.onunhandledrejection, EventTarget.addEventListener, beforeunload/unload/blur                                                                                                                                    | 4  | Error and promise rejection monitoring, global listener tracking, page lifecycle                                             |
| Total               |                                                                                                                                                                                                                                        | 76 |                                                                                                                              |

in isolation rarely produce the specific action combinations (external script injection, cookie gating, beacon exfiltration) that the behavioral rules target.

The two rule sets are complementary. Static rules provide broad recall (93.6%) at the cost of higher per-file false positives, while behavioral rules provide precise confirmation (0.06% FPR) with lower standalone recall (64.1%). The pipeline combines both scores at the cluster level, where static scores flag candidates broadly and behavioral scores confirm them precisely.

Setting every rule to equal weight (equal-weights row in Table 5) reduces static recall from 93.6% to 43.6% and collapses behavioral recall from 64.1% to 7.7%, because high-specificity indicators (e.g., `net_suspicious_domain` at 0% FP and 39.7% TP) are drowned out by frequently firing low-specificity rules. This confirms that the weight calibration reflects empirical specificity rather than arbitrary assignment.

**Table 5: Heuristic evaluation against 5,000 unique human-verified benign JavaScript files and 61 confirmed malicious campaign payloads. Static rules operate on source code and behavioral rules operate on the execution trace of the same file. Zero-FP counts rules that fire on zero benign files.  $FP < 1\%$  counts rules firing on fewer than 1% of benign files. Study weights are the specificity-calibrated weights assigned during the study. Equal weights sets every rule to weight 1.0, removing all differentiation.**

| Metric                 | Static (37) | Behavioral (33) |
|------------------------|-------------|-----------------|
| Zero-FP rules          | 7/37        | 26/33           |
| FP < 1% rules          | 18/37       | 30/33           |
| Benign score median    | 3.5         | 0.0             |
| Benign score P95       | 22.0        | 1.5             |
| Malicious score median | 16.8        | 7.0             |
| Recall (study weights) | 93.6%       | 64.1%           |
| FPR (study weights)    | 22.7%       | 0.06%           |
| Recall (equal weights) | 43.6%       | 7.7%            |

**Table 4: Catalog of the 20 canonical campaigns ordered by approximate harm severity. *Indicators*: fingerprintable artifacts extracted by the pipeline. *Behavior*: effect on victims and site operators.**

| Campaign                     | Indicators                                                                                                                                                                                                                                                                               | Behavior and harm                                                                                                                                                                                                                                                                                        |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WooBackdoor                  | obfuscator.io + RC4 inner stage; 75 KB payload; hidden admin account wooconmercel; WooCommerce REST API calls                                                                                                                                                                            | Installs a persistent administrative backdoor on WooCommerce stores. Creates a hidden admin, exfiltrates customer records, order history, and payment metadata, and hands the attacker long-term control of the store                                                                                    |
| PolyClickFix                 | Base64+XOR payload; Polygon smart contract 0xb36482fe...1a43; 15 fallback RPC endpoints; fake plugin dir file-updater-7b52e912; EtherHiding/ClearFake-style blockchain payload delivery on Polygon rather than BSC [? ? ]; ClickFix lure documented by GoDaddy [? ] and Trend Micro [? ] | Presents visitors with a fake browser update, counterfeit reCAPTCHA, Blue Screen of Death, or Cloudflare interstitial that coerces them into pasting an attacker-supplied PowerShell command into the Windows Run dialog, delivering an infostealer or remote-access trojan to the visitor's own machine |
| MultiC2                      | Loader injected into existing plugin files such as jquery.blockUI.min.js; six fallback C2 domains (goveanrs.org, govearali.org, ligovera.shop, alianzeg.shop, getalia.org, ztdaliweb.shop); recursive tryInfect(); DoubleDonut delivery network [? ]                                     | Acts as a loader for the DoubleDonut delivery network, feeding the same ClickFix PowerShell-paste lure used by PolyClickFix. Because the loader hides inside a legitimate plugin file, it survives cleanups that remove only newly added assets                                                          |
| MalvertKit                   | 171 KB obfuscated payload; navigator.permissions, WebGL, font probe, localStorage persistence; fetch exfiltration to czcf.php and /ad/visit*; hidden iframes; scroll and mouse observers; pop-under windows                                                                              | Fingerprints the visitor across canvas, WebGL, and font channels, exfiltrates the profile, and opens pop-under windows onto attacker-controlled advertising landing pages. Used to push fake-giveaway scams and browser-notification permission prompts that harvest push-delivery rights for later spam |
| JSFiretruck                  | _/static/js/ad.js; JSFiretruck encoding using only []()!+ [? ]; Function constructor                                                                                                                                                                                                     | Silently redirects visitors from the requested page to an attacker-controlled destination, typically a counterfeit landing page or affiliate-fraud scheme. Per-wave encoding defeats hash blocklists                                                                                                     |
| SessHijack-A/B/C             | sessionStorage single-execution gate; CSS page-fade animation; fixed PHP endpoint (e.g., hupe-wa.dz/fifa.php)                                                                                                                                                                            | Hijacks visitor navigation into a server-side traffic distribution service that routes human visitors to phishing portals, fake software-update pages, affiliate-fraud forms, and push-notification permission prompts                                                                                   |
| FadeRedirect                 | @keyframes fadeIn; rotating C2 hosts (e.g., bioredox.cl/pm a.php, primetimehost.me/ama.php)                                                                                                                                                                                              | Same traffic-hijack behavior as SessHijack but rotates C2 hosts across observations so that host-based blocklists expire quickly                                                                                                                                                                         |
| TDSRedirect                  | Loader referencing cdnjslibraries.com, a look-alike CDN domain; consistent with DollyWay [? ] and Parrot TDS [? ]                                                                                                                                                                        | Front-end to an upstream traffic distribution service that forwards visitors to whichever second stage the operator selects at request time, including browser exploit kits, fake updates, and tech-support scam pages                                                                                   |
| YWXLloader                   | Four-line immediately-invoked function expression loading cdn.ywxi.net/js/1.js; injection-point polymorphism (33 unique diff hashes)                                                                                                                                                     | Grants the attacker an on-demand delivery channel anchored at a third-party CDN. Arbitrary JavaScript can be pushed into every victim page at any future time without modifying the compromised site                                                                                                     |
| TagInject-A/B                | Loader at 88/tag.min.js; heavily obfuscated secondary payload from attacker domains such as 6opo.com; naming mimics commercial tag-management libraries                                                                                                                                  | Persistent external-payload foothold disguised as a marketing or analytics tag. The delivered payload can be repointed at any time, allowing the attacker to rotate between redirects, fingerprinting, and malware delivery                                                                              |
| CSSInject                    | Newly created file at api/css.js; runtime CSS-rule injection rather than conventional script behavior                                                                                                                                                                                    | Manipulates layout and element visibility to create space for unauthorized overlay content or to suppress warning banners while a concurrent injection stage executes                                                                                                                                    |
| HexArray-A/B/C<br>GDPRInject | Ad-serving script at ba298f04.js on aj1559.online; MD5 and Base64 fingerprint hashing; <ins> tag scan; postscribe ad rendering; GDPR variant uses asrv.dalecta.com                                                                                                                       | Displaces the site's legitimate ad inventory with fraudulent impressions the attacker is paid for. The site operator loses advertising revenue and visitors receive unvetted ads, a channel prior threat intelligence has linked to second-stage malvertising                                            |
| CookieLoader                 | Cookie-gated activation; waits for marker cookie before executing second stage                                                                                                                                                                                                           | Deferred redirect loader whose activation gate suppresses execution during analysis and triggers only on returning visitors that carry the marker cookie                                                                                                                                                 |
| OverlayInject                | DOM overlay element with attacker-controlled content                                                                                                                                                                                                                                     | Covers portions of the victim page with content the visitor did not request, most commonly used to stage click-hijack or scam overlays                                                                                                                                                                   |

E VirusTotal Detection Timeline

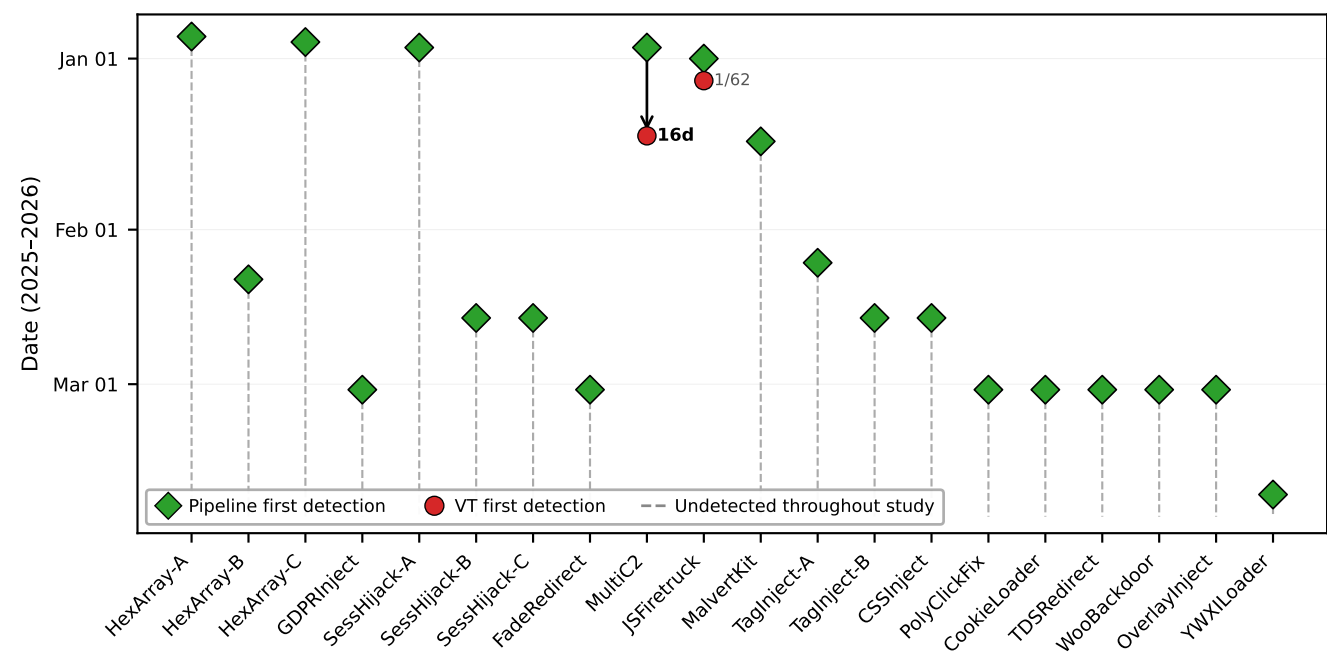


Figure 1: VirusTotal detection gap. Diamonds (◇): pipeline first detection; circles (○): VT first non-zero detection. Dashed lines: undetected at study end. Only MultiC2 received meaningful coverage (23/61 engines, 16-day delay).